UiA Universitetet i Agder

MAS508-G
Lab 3 – System modeling and linearization



Tasks

- **Find the equilibrium point(s).**
- **Linearize the system about the specified equilibrium.**
- **Define the given input/output and derive the transfer function of the linearized model.**
- **Discuss stability from the transfer function poles.**

Task 1: Pendulum with Nonlinear Friction (torque input)

A pendulum of length L, mass m, with damping proportional to $\sin(\dot{\theta})$ is driven by an external torque $\tau(t)$:

$$mL^2\ddot{\theta} + b\,\sin(\dot{\theta}) + mgL\sin(\theta) = \tau(t).$$

I/O definition: input $u(t) = \tau(t)$, output $y(t) = \theta(t)$.

1. Identify the equilibrium (use $\theta_e = 0$, $\dot{\theta}_e = 0$, $u_e = 0$).
2. Linearize about $(\theta_e, \dot{\theta}_e) = (0, 0)$.
3. Find $G_1(s) = \Theta(s)/\tau(s)$.
4. From the poles of $G_1(s)$, discuss stability.

Task 2: Nonlinear Spring–Mass (force input)

A mass–spring with cubic stiffness is driven by a force $F(t)$:

$$m\ddot{x} + k_1 x + k_2 x^3 = F(t).$$

I/O definition: input $u(t) = F(t)$, output $y(t) = x(t)$.

1. Find equilibrium point(s). Consider linearization at $x_e = 0$, $u_e = 0$.
2. Linearize about $x_e = 0$.
3. Find $G_2(s) = X(s)/F(s)$.
4. Use the poles of $G_2(s)$ to assess stability.

Task 3: Nonlinear Two-State System

$$\dot{x}_1 = -x_2 + x_1^2 + u, \qquad \dot{x}_2 = x_1 - x_2^3, \qquad y = x_1.$$

I/O definition: input $u(t)$ enters the $\dot{x}_1$ equation additively; output $y(t) = x_1(t)$.

1. Find all equilibrium points.
2. Linearize at each equilibrium to obtain $\dot{\delta x} = A\delta x + B\delta u,\ y = C\delta x$.
3. Derive the transfer function $G_3(s) = Y(s)/U(s) = C(sI - A)^{-1}B$.
4. Discuss stability from the poles of $G_3(s)$ at each equilibrium.